



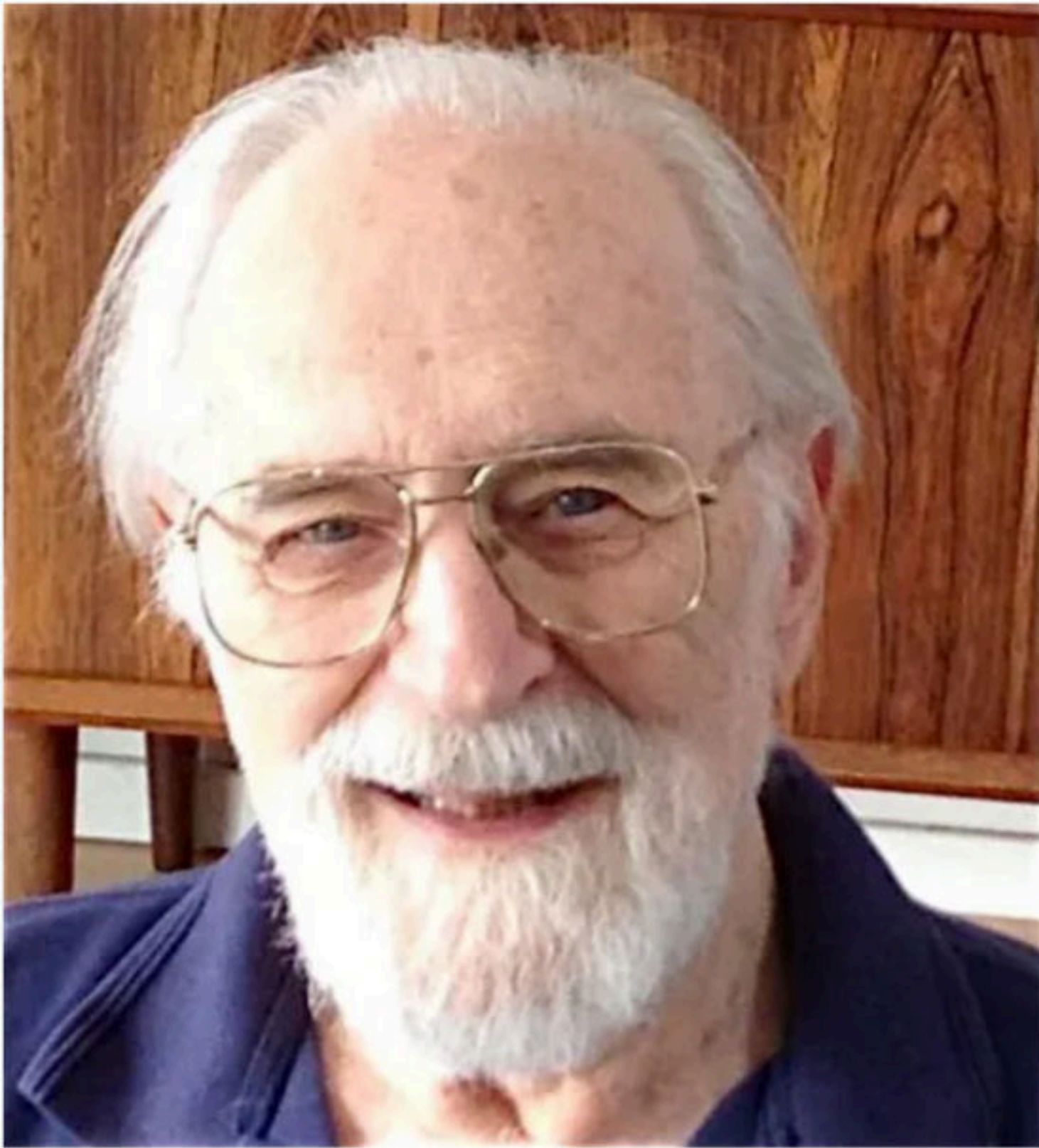
Simplification of Boolean Algebra

Part - III

Foundation Course on Digital Electronics

Simplification of Boolean expression

- After deriving a Boolean expression from the truth table next important step is to minimise it, so that the cost of implementation into hardware and delay because of additional gates can be reduced. There are following methods for simplifying a Boolean expression
 - 1) Using Karnaugh-map
 - 2) Using Algebraic method (Boolean Laws)



- Maurice Karnaugh (born 4 October 1924) is an American physicist, mathematician and inventor known for the Karnaugh map used in Boolean algebra.
- Age: 96

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Karnaugh Map

- **Problem with other methods:** - The algebraic procedure using Boolean laws and rules for minimising Boolean expression becomes difficult when a function becomes complex, because we have to identify where is the scope of minimization based on some Boolean laws and It is difficult to understand where we reach saturation point or not.

$$a + a'b + a'b'c + a'b'c'd'$$

- Karnaugh map is one of the most extensively used tool, it is a graphical representation, represents truth table by pictorial form, provides a systematic method for simplifying or minimizing a Boolean expression. For a n-variable k-map, there will be 2^n cells addressed by a gray code. Each cell corresponds to one minterm or maxterm.

	a	b	c	d	F
0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	
3	0	0	1	1	
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	
7	0	1	1	1	
8	1	0	0	0	
9	1	0	0	1	
10	1	0	1	0	
11	1	0	1	1	
12	1	1	0	0	
13	1	1	0	1	
14	1	1	1	0	
15	1	1	1	1	

	ab				
cd					

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00				
c'd	01				
cd	11				
cd'	10				

		0	4	12	8
	1		5	13	9
	3		7	15	11
	2		6	14	10

	cd				
ab					

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00				
		0	4	12	8
c'd	01				
		1	5	13	9
cd	11				
		3	7	15	11
cd'	10				
		2	6	14	10

	cd	c'd'	c'd	cd	cd'
ab		00	01	11	10
a'b'	00				
		0	1	3	2
a'b	01				
		4	5	7	6
ab	11				
		12	13	15	14
ab'	10				
		8	9	11	10

	ad				
bc					

	db				
ca					

	ad	a'd'	a'd	ad	ad'
bc		00	01	11	10
b'c'	00				
		0	1	9	8
b'c	01				
		2	3	11	10
bc	11				
		6	7	15	14
bc'	10				
		4	5	13	12

	db	d'b'	d'b	db	db'
ca		00	01	11	10
c'a'	00				
		0	4	5	1
c'a	01				
		8	12	13	9
ca	11				
		10	14	15	11
ca'	10				
		2	6	7	3

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	ab				
c					

	bc				
a					

	a		
bc			

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	0	0	2	6	4
c	1	1	3	7	5

	bc	b'c'	b'c	bc	bc'
a		00	01	11	10
a'	0	0	1	3	2
a	1	4	5	7	6

	a	a'	a
bc		0	1
b'c'	00	0	4
b'c	01	1	5
bc	11	3	7
bc'	10	2	6

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	a		
b			

	b		
a			

	a	a'	a
b		0	1
b'	0		
		0	2
b	1		
		1	3

	a	a'	a
b		0	1
b'	0		
		0	2
b	1		
		1	3

Break

K-Map for POS

	$a+b$				
$c+d$					

K-Map for POS

	$a+b$	$a+b$	$a+b'$	$a'+b'$	$a'+b$
$c+d$		$0+0$	$0+1$	$1+1$	$1+0$
$c+d$	$0+0$	0	4	12	8
$c+d'$	$0+1$	1	5	13	9
$c'+d'$	$1+1$	3	7	15	11
$c'+d$	$1+0$	2	6	14	10

Break

- **Minimal function**: - A function is said to be minimal if it is representing the function expression, if it is using minimal no of literals to represent the function.

- Rules of grouping: -

1. Every minterm must be covered
2. Group must have contiguous Cells(circular)
3. Group must be in horizontal or vertical fashion(circular)
4. Number of cells in a group must be in power of 2
5. Will Try to make largest group possible, so that number of literals in the expression can be reduced
6. Can also take don't care if it helps in creating the larger groups, other wise don't care.
7. Will consider new implicant if it is covering some new minterm.

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	0	4	12	8
c'd	01	1	5	13	9
cd	11	3	7	15	11
cd'	10	2	6	14	10

Don't care condition

- Don't care cases are those cases which can never occur logically in that function.
- It is not essential to cover don't care conditions, but if don't care are helping to generate bigger prime implicants, then we can use them.

Break

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00			1	
c'd	01	D	D	1	D
cd	11		1	1	
cd'	10		1	1	

→ $\cancel{bc} + \cancel{ab}$

→ ~~$b(c+a)$~~

→ 4

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00		1		
c'd	01		1	1	1
cd	11	1	1	1	
cd'	10			1	

$\bar{a} b \bar{c}$

+

$a \bar{c} d$

$\bar{a} c d$

+

$a b c$

+

12

First try to cover those minterms which do not have an option.

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Break

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00		D		D
c'd	01	①	1	D	
cd	11		D	1	1
cd'	10		1	D	

$\overline{a}\overline{b}\overline{c}d$ + implicant
 $\overline{a}\overline{c}d$
 prime implicant

EPI



$$\overline{a}\overline{c}d + acd + \frac{\overline{a}b}{bc}$$

① $\overline{a}\overline{c}d + acd + \underline{\underline{a}}b$ ←

② $\overline{a}\overline{c}d + acd + bc$ ←

For some functions more than one minimal Boolean expression are possible

.

Break

- **Implicants**: - Collection of adjacent minterms are called implicants.
- **Prime Implicant (PI)**: - Implicant is called Prime implicant (PI) if it is not subset of any other implicant(group). (overlapping is allowed).
 - We will always try to find Prime Implicant

$$f(a, b, c, d) = \sum_m \{4, 5, 6, 7, 8, 9, 10, 11, 13, 14\}$$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	0	4	12	8
c'd	01	1	5	13	9
cd	11	3	7	15	11
cd'	10	2	6	14	10

$$\bar{a}b + a\bar{b} + \bar{b}c\bar{d} + \bar{a}\bar{c}d + b\bar{c}d + \bar{a}cd$$

Essential Prime Implicant (EPI)

- If a prime implicant (PI) have some unique minterm which no other prime implicant covers then, it is called essential prime implicant (EPI).
- Essential prime implicant (EPI) must always be present in the minimal Boolean expression.

Break

Q $f(a, b, c) = \sum_m \{1, 2, 3, 4, 5\}$, find Number of PI, Number of EPI, Number of literals are there in minimal expression, Number of different minimal expression possible?

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	0		1		1
c	1	1	1		1

Diagram illustrating the Karnaugh map for the function $f(a, b, c) = \sum_m \{1, 2, 3, 4, 5\}$. The map shows the following groupings:

- Group 1 (Red box): $a'b$ (minterms 2 and 3)
- Group 2 (Blue box): $a'b'$ (minterms 0 and 1)
- Group 3 (Orange box): ab' (minterms 4 and 5)
- Group 4 (Purple circle): c (minterms 1, 3, 4, 5)

1. Number of PI (a) $\bar{a}b$ $a\bar{b}$ $\bar{a}c$ $\bar{b}c$
2. Number of EPI $\downarrow \downarrow$
3. Number of different minimal expression possible $\rightarrow 2$
4. Number of literals are there in minimal expression $\rightarrow 6$

Q Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is 3. (GATE-2018) (2 Marks)

	ab	$a'b'$	$a'b$	ab	ab'
cd		00	01	11	10
$c'd'$	00	1 0	4	d ₁₂	d ₈
$c'd$	01	1	1 5	13	1 9
cd	11	d ₃	1 7	15	1 11
cd'	10	2	6	d ₁₄	d ₁₀

0, 2, 8, 10

8 9 10 11

Q The total number of prime implicants of the function $f(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 10)$ is 3. (GATE-2015) (1 Marks)

(A) 2

(B) 3

(C) 4

(D) 5

$$\bar{a}\bar{d} + \bar{a}b\bar{c} + \bar{b}c\bar{d}$$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	1 0	1 4	12	8
c'd	01	1	1 5	13	9
cd	11	3	7	15	11
cd'	10	1 2	1 6	14	10

0 2 4 6

Q Which are the essential prime implicants of the following Boolean function?

(GATE-2004) (1 Marks)

$$f(a, b, c) = a'c + ac' + b'c$$

(A) $a'c$ and ac'

(C) $a'c$ only

(B) $a'c$ and $b'c$

(D) ac' and bc'

	ab	$a'b'$	$a'b$	ab	ab'
c		00	01	11	10
c'	0	0	2	6	4
c	1	1	3	7	5

Break

Q Consider the functions given below ?

	ab	$a'b'$	$a'b$	ab	ab'
c		00	01	11	10
c'	0	1	1		1
c	1		1	1	1

1. Number of PI
2. Number of EPI
3. Number of different minimal expression possible
4. Number of literals are there in minimal expression

Q Consider the functions given below ?

1. Number of PI — 8
2. Number of EPI — 0
3. Number of different minimal expression possible — 2
4. Number of literals are there in minimal expression — 12

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	1			1
c'd	01	1	1		
cd	11		1	1	
cd'	10			1	1

Break

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	1	D	D	1
c'd	01	D			
cd	11				
cd'	10	1			D

$\overline{b} \overline{d}$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00		1	1	
c'd	01	D			1
cd	11	D			1
cd'	10		1	1	D

The Karnaugh map shows the following groupings:

- Red Groupings:**
 - A group of two cells at (cd, a'b) and (cd, ab) with value 1.
 - A group of two cells at (c'd', a'b) and (c'd', ab) with value 1.
 - A group of two cells at (cd', a'b) and (cd', ab) with value 1.
- Purple Groupings:**
 - A group of two cells at (c'd, a'b') and (cd, a'b') with value D.
 - A group of two cells at (cd, ab') and (cd', ab') with value 1.

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	D	1		1
c'd	01		1	D	
cd	11	1	D	D	
cd'	10	D			D

$$\overline{b}d + \overline{a}b\overline{c} + \overline{a}\overline{b}c$$

③

$$p_i = 4$$

$$e.p_i = 2$$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00			D	1
c'd	01		1	D	1
cd	11		1	D	D
cd'	10		1	D	D

$$\phi + bd + bc$$

Q Simplify the following using K-map:

$$F(A, B, C, D) = \sum (0, 1, 2, 8, 9, 12, 13)$$

$d(A, B, C, D) = \sum (10, 11, 14, 15)$ d stands for don't care condition. (NET-JULY-2018)

(a) $A + B' D' + BC$

(b) $A + B' D' + B' C'$

(c) $A' + B' C'$

(d) $A' + B' C' + B' D'$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	0	4	12	8
c'd	01	1	5	13	9
cd	11	3	7	15	11
cd'	10	2	6	14	10

Q The Boolean function with the Karnaugh map (NET-NOV-2017)

(a) $(A+C).D+B$

(b) $(A+B).C+D$

(c) $(A+D).C+B$

(d) $(A+C).B+D$

$\underline{AD} + \underline{CD} + \underline{B}$

AB \ CD		AB			
		00	01	11	10
CD	00	0	1	1	0
	01	0	1	1	1
	11	1	1	1	1
	10	0	1	1	0

Q Given $f(w, x, y, z) = \sum_m(0,1,2,3,7,8,10) + \sum_d(5,6,11,15)$, where d represents the don't-care condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$? (GATE-2017) (2 Marks)

- a) $f = (w' + z')(x' + z)$ b) $f = (w' + z)(x + z)$ c) $f = (w + z)(x' + z)$ d) $f = (w + z')(x' + z)$

	wx	w'x'	w'x	wx	wx'
yz		00	01	11	10
y'z'	00	1 ₀			1 ₈
y'z	01	1 ₁	d ₅		
yz	11	1 ₃	1 ₇	d ₁₅	d ₁₁
yz'	10	1 ₂	d ₆		1 ₁₀

	w+x	w+x	w+x'	w'+x'	w'+x
y+z		0+0	0+1	1+1	1+0
y+z	0+0		0 ₄	0 ₁₂	
y+z'	0+1		d ₅	0 ₁₃	0 ₉
y'+z'	1+1			d ₁₅	d ₁₁
y'+z	1+0		d ₆	0 ₁₄	

$$(\bar{x} + z) \cdot (\bar{w} + \bar{z})$$

→ 30

Q The Karnaugh map for a Boolean function is given as **(NET-AUG-2016)**

The simplified Boolean equation for the above Karnaugh Map is

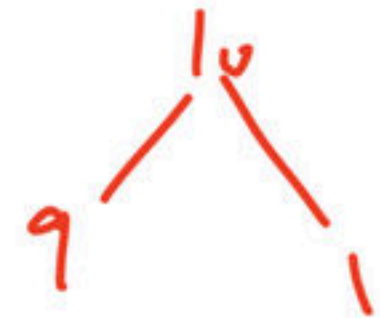
(a) $AB + CD + AB' + AD$

(c) $AB + AD + BC + ACD$

 **(b)** $AB + AC + AD + BCD$

(d) $AB + AC + BC + BCD$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	0	0	1	0
AB	1	1	1	1
$A\bar{B}$	0	1	1	1



Q Consider the following minterm expression for F: **(GATE-2014) (2 Marks)**

$$F(P, Q, R, S) = \Sigma (0, \underline{2}, 5, \underline{7}, \underline{8}, 10, \underline{13}, 15)$$

The minterm 2, 7, 8, 13 are 'do not care' term.

The minimum sum-of-products from for F is

a) $QS' + Q'S$

b) $Q'S' + QS$ — $q \oplus s$

c) $Q'R'S' + Q'RS' + QR'S + QRS$

d) $P'Q'S' + P'QS + PQS + PQ'S'$

	pq	p'q'	p'q	pq	pq'
rs		00	01	11	10
r's'	00	0	4	12	8
r's	01	1	5	13	9
rs	11	3	7	15	11
rs'	10	2	6	14	10

Q The simplified function in product of sums of Boolean function $F(W, X, Y, Z) = \sum(0, 1, 2, 5, 8, 9, 10)$ is (NET-JUNE-2013)

(A) ~~$(W' + X') (Y' + Z') (X' + Z)$~~

(C) $(W' + X') (Y' + Z) (X' + Z)$

(B) $(W' + X') (Y' + Z') (X' + Z')$

(D) $(W' + X') (Y + Z') (X' + Z)$

	wx	w'x'	w'x	wx	wx'
yz		00	01	11	10
y'z'	00	0	4	12	8
y'z	01	1	5	13	9
yz	11	3	7	15	11
yz'	10	2	6	14	10

	w+x	w+x	w+x'	w'+x'	w'+x
y+z		0+0	0+1	1+1	1+0
y+z	0+0	0	4	1	8
y+z'	0+1	1	5	13	9
y'+z	1+1	3	7	15	11
y'+z	1+0	2	6	14	10

Q What is the minimal form of the Karnaugh map shown below? Assume that X denotes a don't care term. **(GATE-2012) (2 Marks)**

(A) $b'd'$

(B) $b'd' + b'c'$

(C) $b'd' + a'b'c'd'$

(D) $b'd' + b'c' + c'd'$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	1	D	D	1
c'd	01	D			1
cd	11				
cd'	10	1			D

Q (NET-JUNE-2010)

The function represented by the k-map given below is



A \ BC				
	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
0	1	0	0	1
1	1	0	0	1



- (A) $A \cdot B$
- (B) $AB + BC + CA$
- (C) $\overline{B \oplus C}$
- (D) $A \cdot B \cdot C$

Q In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map? **(GATE-2008) (2 Marks)**

A) $b'd' + a'd'$

B) $a'b' + b'd' + a'bd'$

C) $b'd' + a'bd'$

D) $a'b' + b'd' + a'd'$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	1	1		1
c'd	01	D			
cd	11	D			
cd'	10	1	1		D

Q Consider the following Boolean function of four variables: **(GATE-2007) (2 Marks)**

$f(w, x, y, z) = \sum (1, 3, 4, 6, 9, 11, 12, 14)$

The function is:

(A) independent of one variable.

(B) independent of two variables. ✓

(C) independent of three variables.

(D) dependent on all the variables.

	wx	w'x'	w'x	wx	wx'
yz		00	01	11	10
y'z'	00	0	4	12	8
y'z	01	1	5	13	9
yz	11	3	7	15	11
yz'	10	2	6	14	10

$$x\bar{z} + \bar{x}z$$

$$x \oplus z$$

Q The switching expression corresponding to $f(A, B, C, D) = \Sigma (1, 4, 5, 9, 11, 12)$ is
(GATE-2005) (2 Marks)

(A) $BC'D' + A'G'D + AB'D$

(B) $ABC' + ACD + B'C'D$

(C) $ACD' + A'BC' + AC'D'$

(D) $A'BD + ACD' + BCD'$

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	0	4	12	8
c'd	01	1	5	13	9
cd	11	3	7	15	11
cd'	10	2	6	14	10

Q The literal count of a Boolean expression is the sum of the number of times each literal appears in the expression. For example, the literal count of $(xy + xz')$ is 4. What are the minimum possible literal counts of the product-of-sum and sum-of-product representations respectively of the function given by the following Karnaugh map? Here, X denotes "don't care" **(GATE-2003) (2 Marks)**

(A) (11, 9)

(B) (9, 13)

(C) (9, 10)

(D) (11, 11)

	zw	z'w'	z'w	zw	zw'
xy		00	01	11	10
x'y'	00	X	1	0	1
x'y	01	0	1	X	0
xy	11	1	X	X	0
xy'	10	X	0	0	X

POS / SOP
 9, $\frac{3+3+2}{8}$
 2+2+2+3

Q Minimum sum of product expression for $f(w, x, y, z)$ shown in Karnaugh-map below is **(GATE-2002) (2 Marks)**

(A) $xz + y'z$

(B) $xz' + zx'$

(C) $x'y + zx'$

(D) ~~None of these~~

	ab	a'b'	a'b	ab	ab'
cd		00	01	11	10
c'd'	00	D	1	0	1
c'd	01	0	1	D	0
cd	11	1	D	D	0
cd'	10	D	0	0	D

Q Given the following Karnaugh map, which one of the following represents the minimal Sum-Of-Products of the map? **(GATE-2001) (2 Marks)**

(A) $xy + y'z$

(B) $wx'y' + xy + xz$

(C) $w'x + y'z + xy$

(D) $xz + y$

	wx	w'x'	w'x	wx	wx'
yz		00	01	11	10
y'z'	00	0	D	0	D
y'z	01	D	1	D	1
yz	11	0	D	1	0
yz'	10	0	1	D	0

$\bar{y}z$

xy

NOT

Q Let $f(w, x, y, z) = \sum (0, 4, 5, 7, 8, 9, 13, 15)$. Which of the following expressions are NOT equivalent to f ? **(GATE-2007) (2 Marks)**

(A) $x'y'z' + w'xy' + wy'z + xz$

(B) $w'y'z' + wx'y' + xz$

(C) $w'y'z' + wx'y' + xyz + xy'z$

(D) $x'y'z' + wx'y' + w'y$

	wx	w'x'	w'x	wx	wx'
yz		00	01	11	10
y'z'	00	0	4	12	8
y'z	01	1	5	13	9
yz	11	3	7	15	11
yz'	10	2	6	14	10

Q Which function does NOT implement the Karnaugh map given below? (GATE - 2000) (2 Marks)

A) $(W + X) Y$

B) $XY + YW$

C) $(W + X) (W' + Y) (X' + Y)$

D) none of the above

	wz	w'z'	w'z	wz	wz'
xy		00	01	11	10
x'y'	00	0	X	0	0
x'y	01	0	X	1	1
xy	11	1	1	1	1
xy'	10	0	X	0	0

DATA

Q The simplified form of a Boolean equation $(AB' + AB'C + AC)(A'C' + B')$ is: (NET-JULY-2016)

(A) $A B'$

(B) $AB'C$

(C) $A'B$

(D) ABC

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	00	0	2	6	4
c	01	1	3	7	5

	a+b	a+b	a+b'	a'+b'	a'+b
C		0+0	0+1	1+1	1+0
C'	0	0	2	6	4
C	1	1	3	7	5

$$F(P, Q, R, S) = PQ + P'QR + P'QR'S$$

a) $PQ + QR + QS$

b) $P + Q + R + S$

c) $P' + Q' + R' + S'$

d) $P'R + P'R'S + P$

	pq	$p'q'$	$p'q$	pq	pq'
rs		00	01	11	10
$r's'$	00	0	4	12	8
$r's$	01	1	5	13	9
rs	11	3	7	15	11
rs'	10	2	6	14	10

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Q The sum of products expansion for the function **(NET-DEC-2013)**

$F(x, y, z) = (x + y)z'$ is given as

(A) $x' y' z + x y z' + x' y z'$

(C) $x y' z' + x' y' z' + x y z'$

(B) $x y z + x y z' + x y' z'$

(D) $x y z' + x y' z' + x' y z'$

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00	0	2	6	4
z	01	1	3	7	5

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Q The simplified SOP (Sum of Product) form of the Boolean expression $(P + Q' + R'). (P + Q' + R). (P + Q + R')$ is **(GATE-2011)(1 Marks)**

(A) $(P'.Q + R')$

(B) $(P + Q'.R')$

(C) $(P'.Q + R)$

(D) $(P.Q + R)$

	pq	$p'q'$	$p'q$	pq	pq'
rs		00	01	11	10
$r's'$	00	0	2	6	4
$r's$	01	1	3	7	5

	$p+q$	$p+q$	$p+q'$	$p'+q'$	$p'+q$
r		0+0	0+1	1+1	1+0
r	0	0	2	6	4
r'	1	1	3	7	5

Q (NET-JUNE-2010)

$AB + \overline{(A + B)}$ is equivalent to

- (A) $A \oplus B$
- (B) $A \odot B$
- (C) $(A \oplus B) \odot A$
- (D) $(A \odot B) \oplus A$

	a	a'	a
b		0	1
b'	0	0	2
b	1	1	3

Q (NET-DEC-2009)

The Boolean expression $\bar{x} \bar{y} z + yz + xz$ is equivalent to :

- (A) x (B) y (C) z (D) $x + y + z$

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00	0	2	6	4
z	01	1	3	7	5

Q The simplified form of the Boolean expression $(X + Y + XY)(X + Z)$ is
(NET-DEC-2009)

(A) $X + Y + ZX + Y$

(C) $X + YZ$

(B) $XY - YZ$

(D) $XZ + Y$

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00	0	2	6	4
z	01	1	3	7	5

Q If P, Q, R are Boolean variables, then $(P + Q')(PQ' + PR)(P'R' + Q')$ simplifies **(GATE-2008) (1 Marks)**

(A) PQ'

	pq	$p'q'$	$p'q$	pq	pq'
rs		00	01	11	10
$r's'$	00	0	2	6	4
$r's$	01	1	3	7	5

(B) PR'

(C) $PQ' + R$

(D) $PR' + Q$

	$p+q$	$p+q$	$p+q'$	$p'+q'$	$p'+q$
r		0+0	0+1	1+1	1+0
r	0	0	2	6	4
r'	1	1	3	7	5

Q Simplified form of Boolean expression $xy + (\sim x)z + yz$ is: **(NET-DEC-2007)**

(A) $xy + (\sim x)z$

(C) $(\sim x)y + xz$

(B) $(\sim x)y + (\sim x)z$

(D) $xy + xz$

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00	0	2	6	4
z	01	1	3	7	5

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Q The logic expression $x'yz' + x'yz + xyz' + xyz$ reduces to: (NET-JUNE-2005)

(A) $x'z$

(B) xyz

(C) y

(D) yz

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00	0	2	6	4
z	01	1	3	7	5

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Q The function $AB'C + A'BC + ABC' + A'B'C + AB'C'$ is equivalent to (GATE-2004) (1 Marks)

(A) $AC' + AB + A'C$

(B) $AB' + AC' + A'C$

(C) $A'B + AC' + AB'$

(D) $A'B + AC + AB'$

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	00	0	2	6	4
c	01	1	3	7	5

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Q Which of the following expressions is equivalent to $(A \oplus B) \oplus C$ (GATE-2004) (2 Marks)

(A) $(A+B+C)(A'+B'+C')$

(B) $(A+B+C)(A'+B'+C)$

(C) $ABC + A'(B \oplus C) + B'(A \oplus C)$

(D) None

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	00				
		0	2	6	4
c	01				
		1	3	7	5

Q The Boolean function $x'y' + xy + x'y$ is equivalent to (GATE-2004) (1 Marks)

- (A) $x' + y'$
- (B) $x + y$
- (C) $x + y'$
- (D) $x' + y$

	x	x'	X
y		0	1
y'	0	02	
y	1	13	

Application

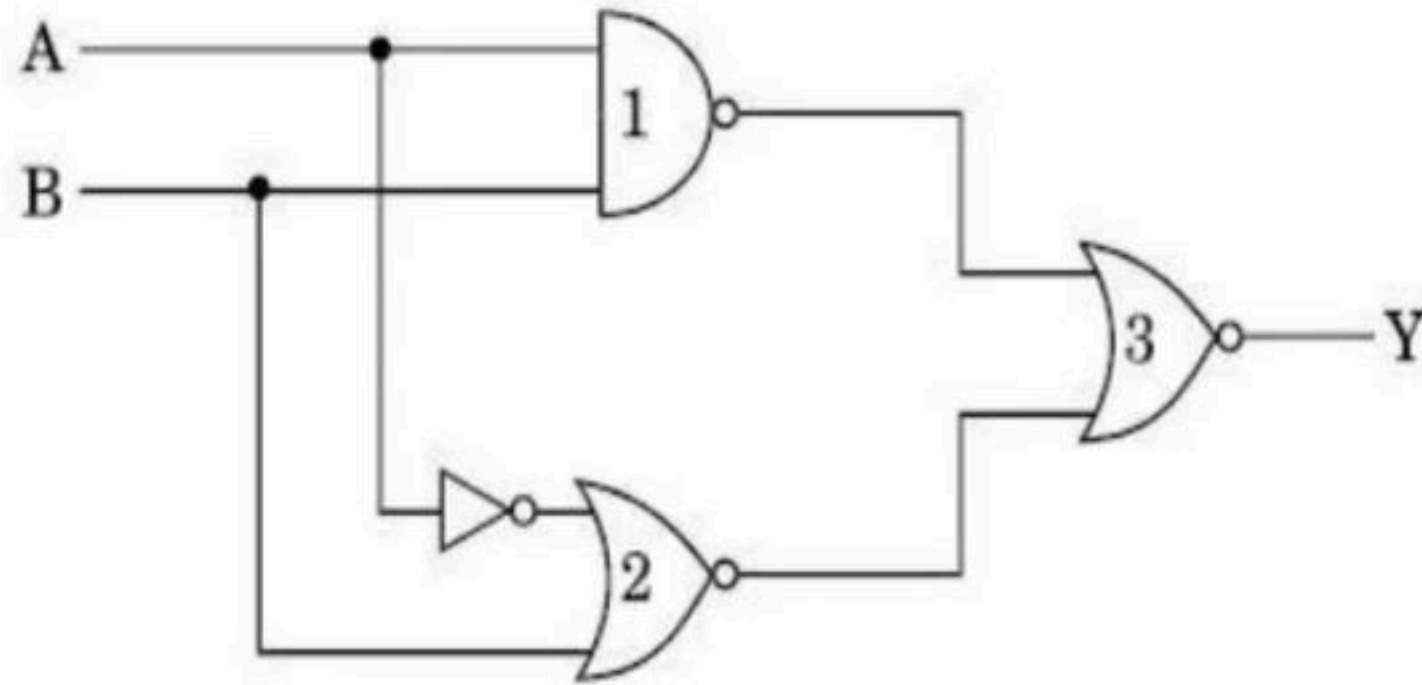
Q find the Boolean expression for the logic circuit shown below: **(NET-DEC-2018)**

a) ab'

b) $a'b$

c) ab

d) $a'b$



Q If w, x, y, z are Boolean variables, then which one of the following is INCORRECT? **(GATE-2017)**
(2 Marks)

a) $wx + w(x + y) + x(x + y) = x + wy$

b) $(wx'(y + z'))' + w'x = w' + x + y'z$

c) $(wx'(y + xz') + w'x')y = xy'$

d) $(w + y)(wxy + wyz) = wxy + wyz$

	wx	$w'x'$	$w'x$	wx	wx'
yz		00	01	11	10
$y'z'$	00	0	4	12	8
$y'z$	01	1	5	13	9
yz	11	3	7	15	11
yz'	10	2	6	14	10

Q The output of the following combinational circuit is F. **(NET-NOV-2017)**

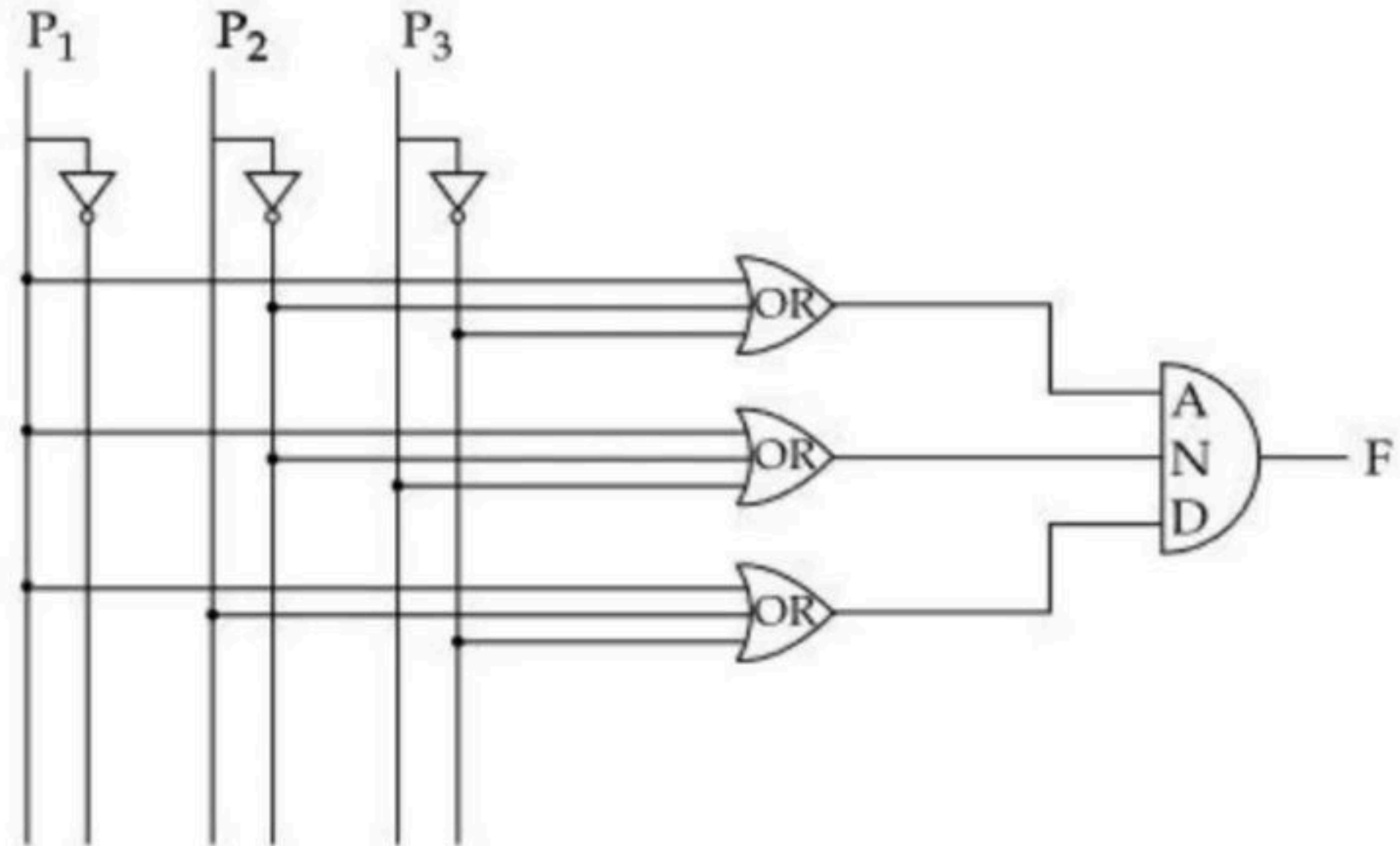
The value of F is :

(a) $P_1 + P_2' P_3$

(b) $P_1 + P_2' P_3'$

(c) $P_1 + P_2 P_3'$

(d) $P_1' + P_2 P_3$



Q Simplified Boolean equation for the following truth table is: **(NET-JULY-2016)**

(a) $F = yz' + y'z$

(b) $F = xy' + x'y$

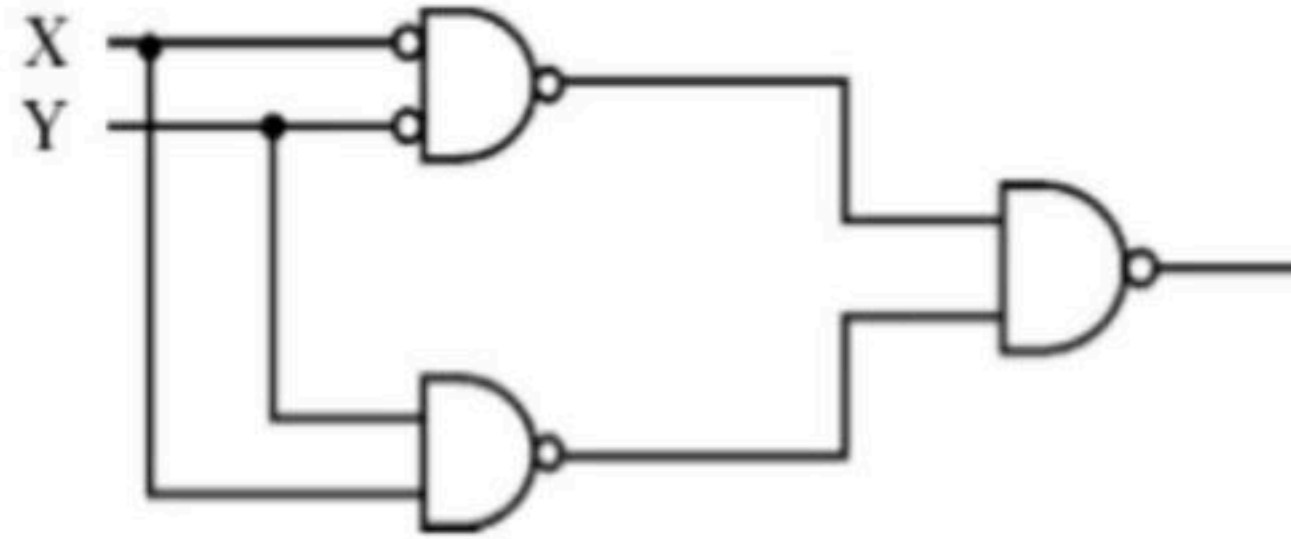
(c) $F = x'z + xz'$

(d) $F = x'z + xz' + xyz$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

	xy	x'y'	x'y	xy	xy'
z		00	01	11	10
z'	0	0	2	6	4
z	1	1	3	7	5

Q The output of the following combinational circuit is: **(NET-AUG-2016)**



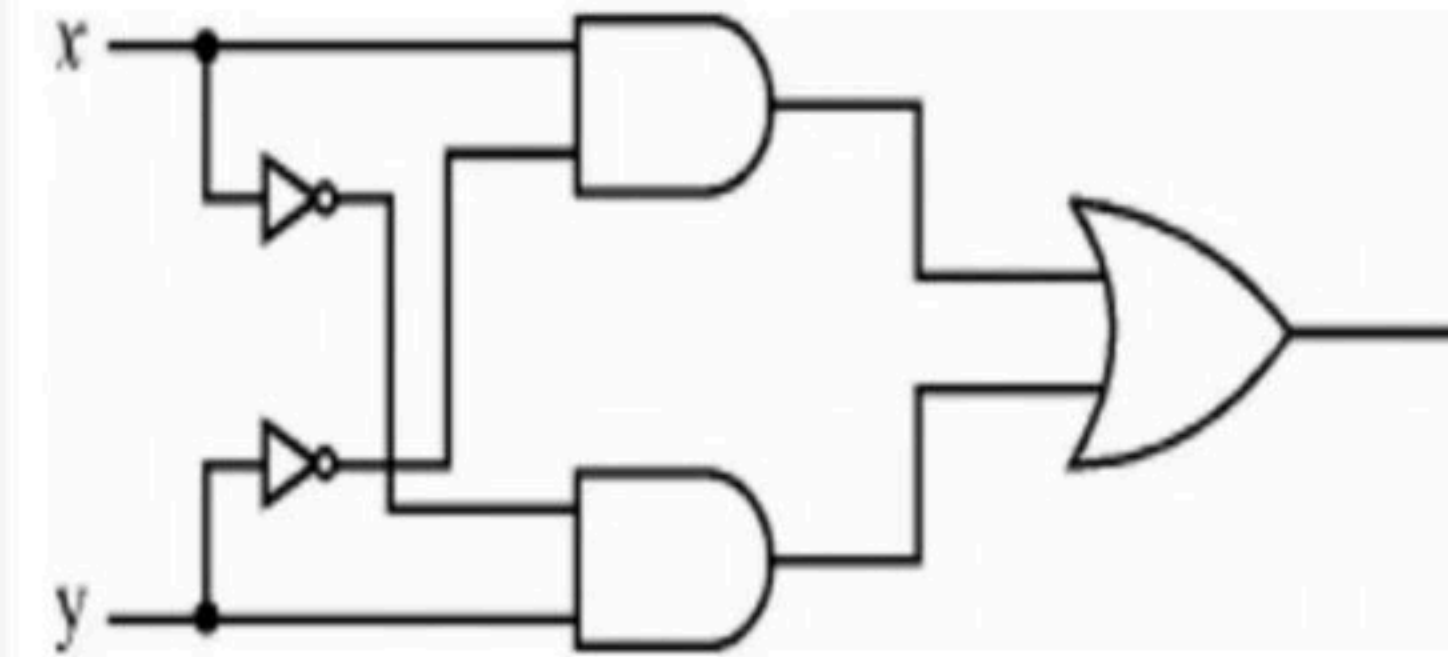
(a) $X \cdot Y$

(b) $X + Y$

(c) $X \oplus Y$

(d) $(X \oplus Y)'$

Q What will be the output of the following logic diagram? (NET-DEC-2013)



	x	x'	x
y		0	1
y'	0	0	2
y	1	1	3

- (A) x OR y
- (B) x AND y
- (C) x XOR y
- (D) x XNOR y

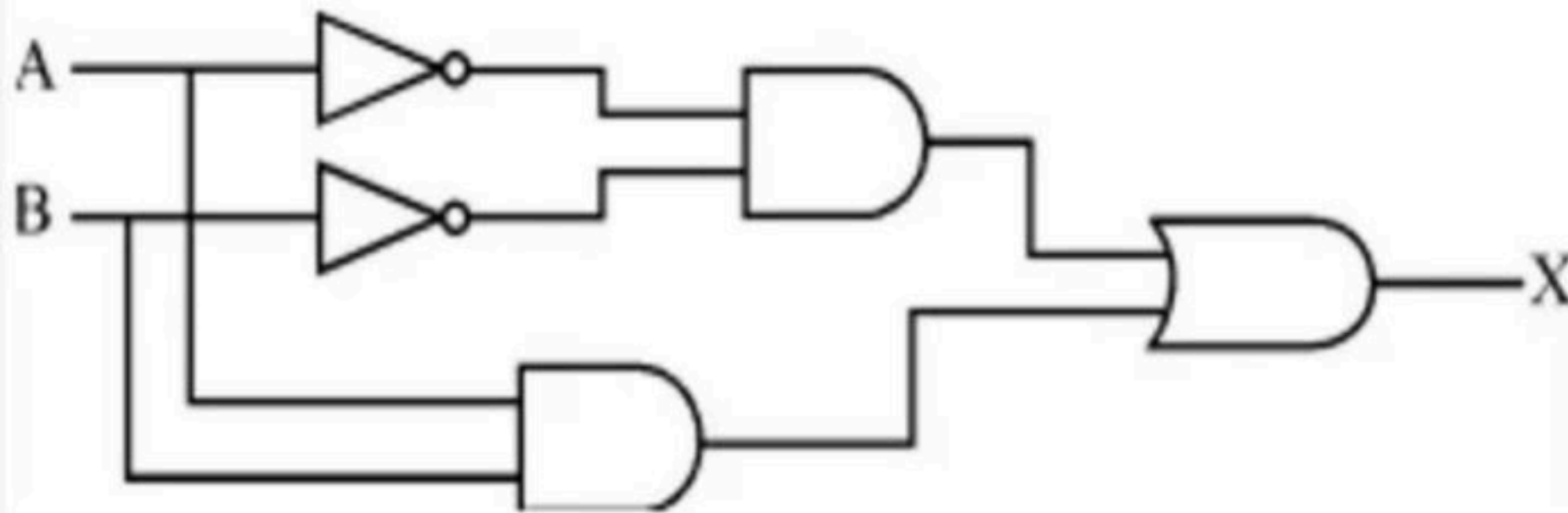
Q What type of logic circuit is represented by the figure shown below? (NET-SEPT-2013)

(A) XOR

(B) XNOR

(C) XAND

(D) XNAND



	a	a'	a
b		0	1
b'	0	0	2
b	1	1	3

Q $(A + B) (AB)'$ is equivalent to (NET-JUNE-2011)

(A) $A \oplus B$

(B) $A \cup B$

(C) $(A \oplus B) \cup A$

(D) $(A \cup B) \oplus A$

Q Let $f(A, B) = A' + B$. Simplified expression for function $f(f(x + y, y), z)$ is : **(GATE-2002) (2 Marks)**

(A) $x' + z$

(B) xyz

(C) $xy' + z$

(D) None of these

Functions

Q Consider two functions f_1 and f_2 ?

$$F_1 = \sum_m (1, 2, 5, 6)$$

$$F_2 = \sum_m (2, 3, 4, 5)$$

Find $f_1.f_2$ and $f_1 + f_2$

Q Consider two functions f_1 and f_2 ?

$$F_1 = \sum_m (1, 2, 5, 6)$$

$$F_2 = \sum_m (2, 3, 4, 5)$$

Find $f_1 \cdot f_2$ and $f_1 + f_2$

	F_1	F_2	$f_1 + f_2$	$f_1 \cdot f_2$
0	0	0		
1	1	0		
2	1	1		
3	0	1		
4	0	1		
5	1	1		
6	1	0		
7	0	0		

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Q Consider f_1 & f_2 –

$$F_1(a, b, c) = \sum_m (0, 2, 4) + d(3, 5, 7)$$

$$F_2(a, b, c) = \sum_m (2, 3) + d(1, 6, 7)$$

Find $f_1 \cdot f_2$ and $f_1 + f_2$

	F_1	F_2	$f_1 + f_2$	$f_1 \cdot f_2$
0	1	0		
1	0	D		
2	1	1		
3	D	1		
4	1	0		
5	D	0		
6	0	D		
7	d	d		

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Q Consider f_1 & f_2 –
 $F_1(a, b, c, d) = \sum_m (1, 3, 4, 5, 9, 10, 11) + d (6, 8)$
 $F_2(a, b, c, d) = \sum_m (0, 2, 4, 7, 8, 15) + d (9, 12)$
 Find $f_1 \cdot f_2$ and $f_1 + f_2$

	F_1	F_2	$f_1 + f_2$	$f_1 \cdot f_2$
0	0	1		
1	1	0		
2	0	1		
3	1	0		
4	1	1		
5	1	0		
6	D	0		
7	0	1		
8	D	1		
9	1	D		
10	1	0		
11	1	0		
12	0	D		
13	0	0		
14	0	0		
15	0	1		

Q Consider a function $f(a, b, c) = \sum_m (3, 5, 6)$ is being minimized to $A + BC$. Predict what are the don't care conditions?

a) $d(2,4)$ **b)** $d(2,7)$ **c)** $d(4,7)$ **d)** $d(2,4,7)$

	ab	a'b'	a'b	ab	ab'
c		00	01	11	10
c'	00	0	2	6	4
c	01	1	3	7	5

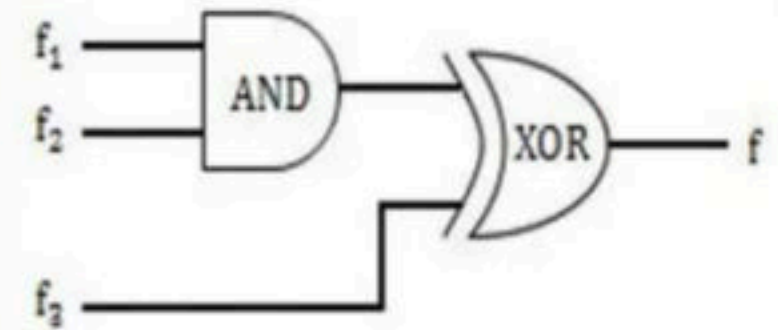
Q Consider three 4-variable functions f_1 , f_2 , and f_3 , which are expressed in sum-of-minterms as

$$f_1 = (0, 2, 5, 8, 14)$$

$$f_2 = (2, 3, 6, 8, 14, 15)$$

$$f_3 = (2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:



For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as: **(GATE-2019) (2 Marks)**

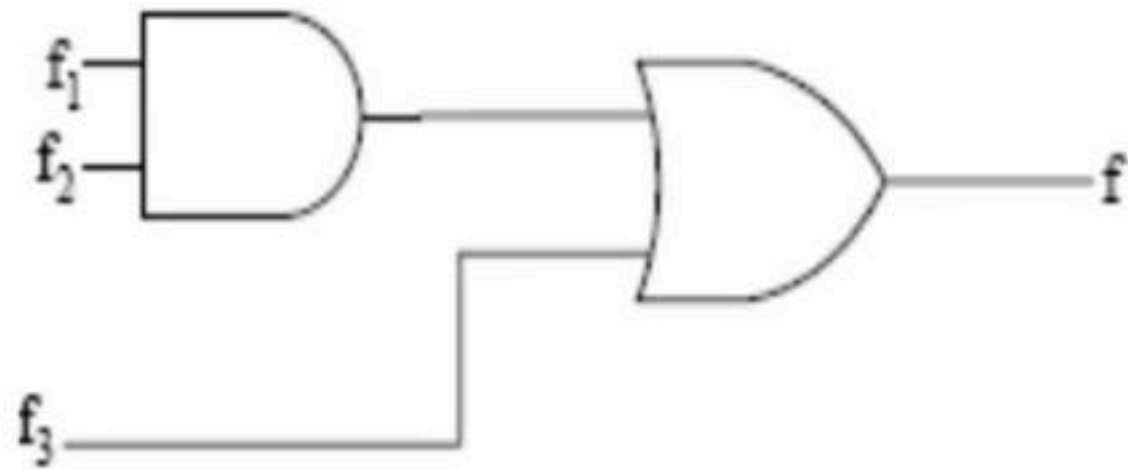
a) $\Sigma(7, 8, 11)$

b) $\Sigma(2, 7, 8, 11, 14)$

c) $\Sigma(2, 14)$

d) $\Sigma(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

Q Given f_1 , f_3 and f in canonical sum of products form (in decimal) for the circuit **(GATE-2008) (1 Marks)**



$$f_1 = \sum m(4, 5, 6, 7, 8)$$

$$f_3 = \sum m(1, 6, 15)$$

$$f = \sum m(1, 6, 8, 15)$$

then f_2 is

A) $\sum m(4, 6)$

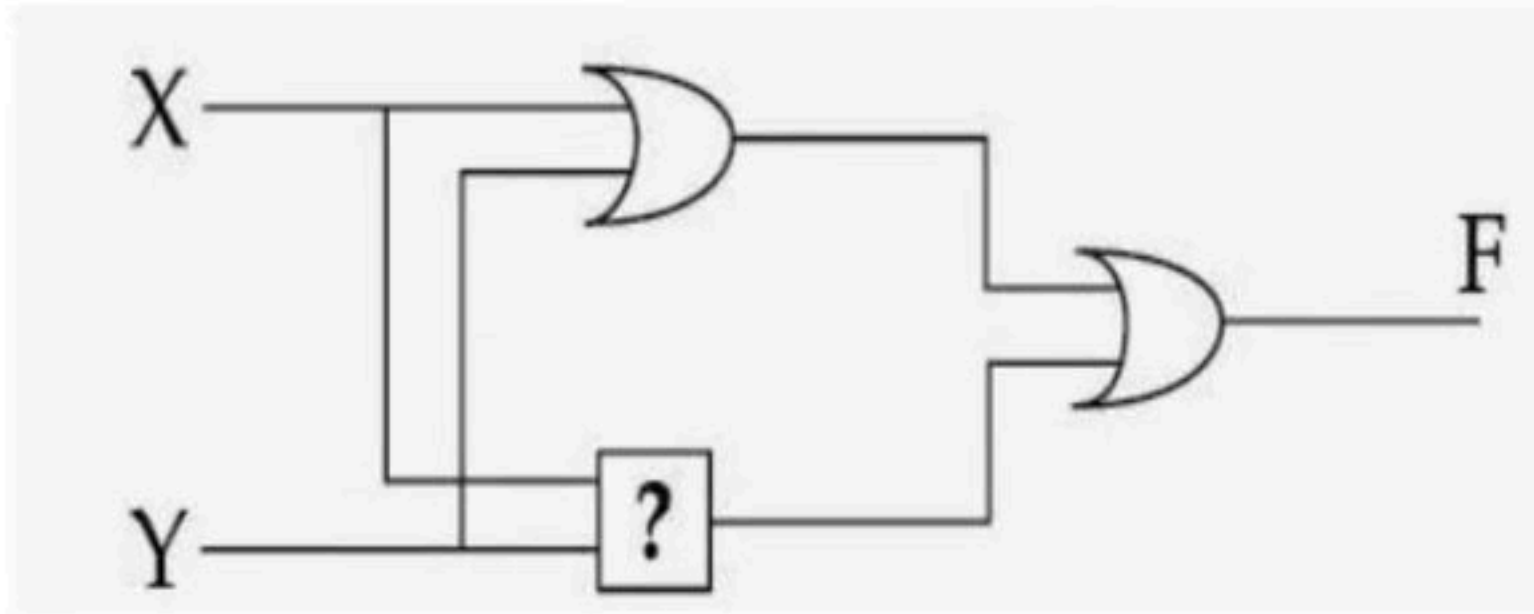
B) $\sum m(4, 8)$

C) $\sum m(6, 8)$

D) $\sum m(4, 6, 8)$

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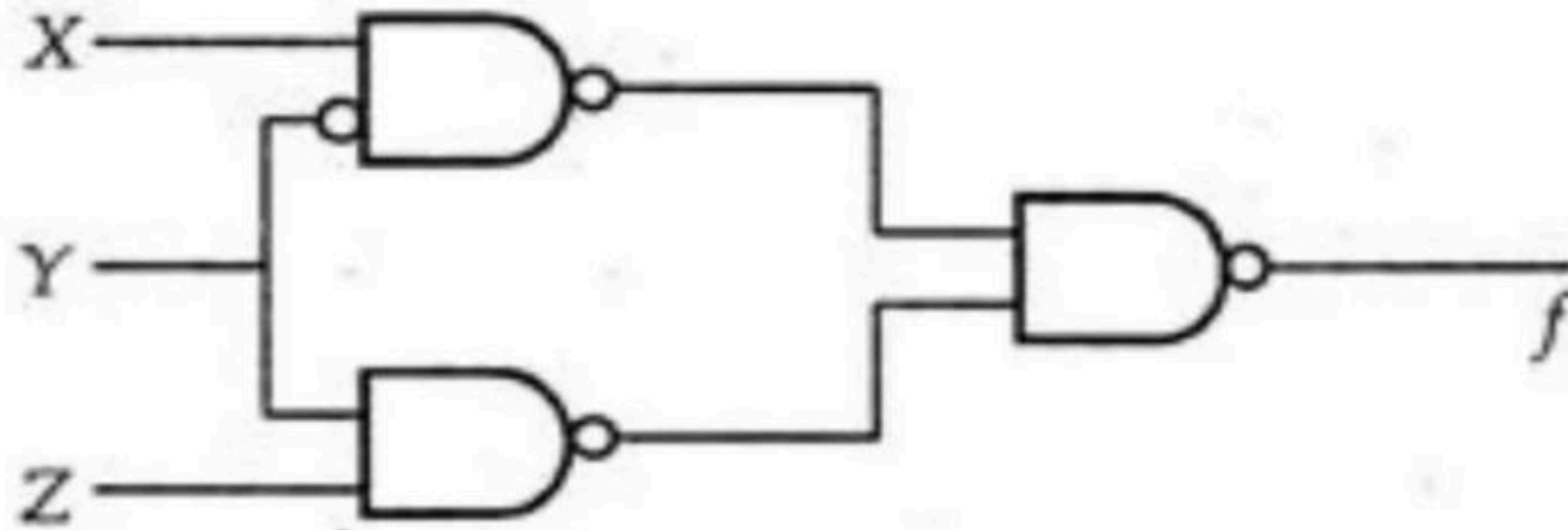
Q Consider the following circuit: **(NET-DEC-2005)**



to make it a Tautology the [?] should be:

- (A)** NAND gate
- (B)** AND gate
- (C)** OR gate
- (D)** EX-OR gate

Q Consider the following circuit. **(GATE-2005) (1 Marks)**
Which one of the following is TRUE?



(A) f is independent of X
(C) f is independent of Z

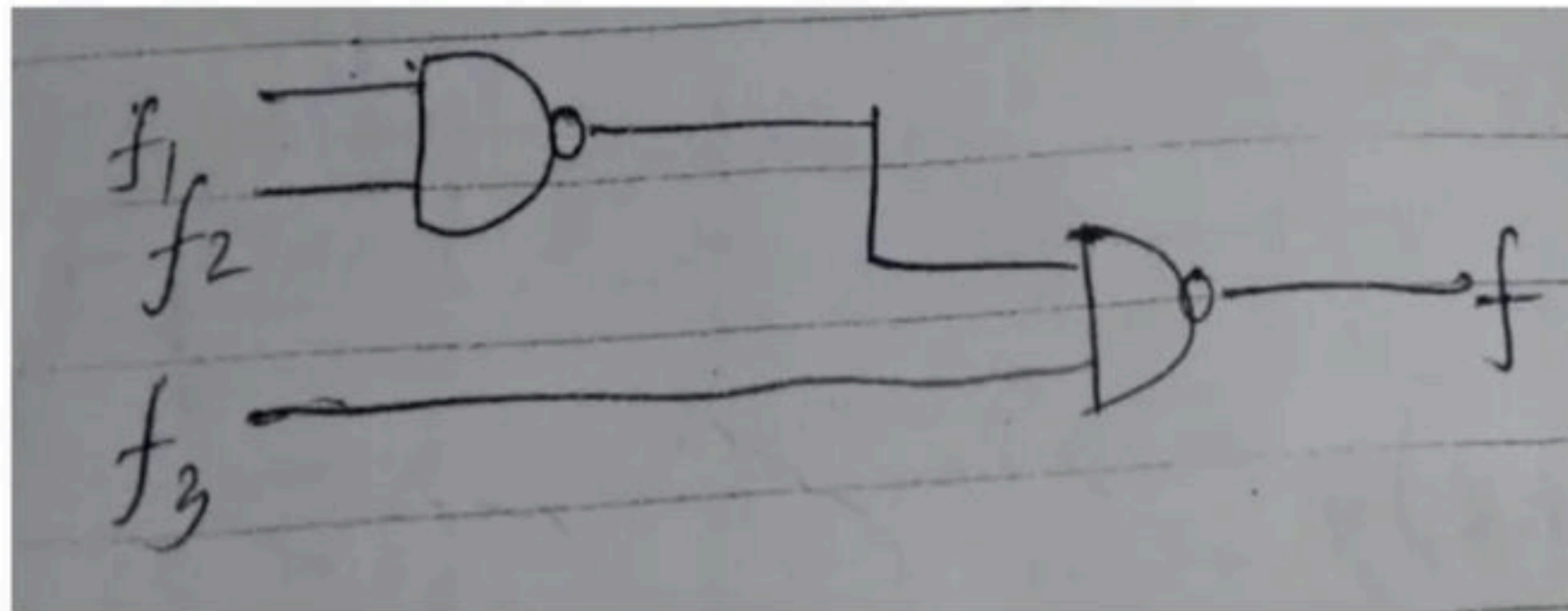
(B) f is independent of Y
(D) None of X, Y, Z is redundant

$$F_1(a, b, c, d) = \sum_m (0, 2, 5, 6, 8, 10, 15)$$

$$F_2(a, b, c, d) = \sum_m (0, 5, 8, 12, 14)$$

$$F(a, b, c, d) = \sum_m (0, 5, 6, 7, 8, 12, 13)$$

$$f_3(a, b, c, d) = ?$$



Break

- **Irredundant function**: - A function f is said to be irredundant if no product term can be removed from the function without changing the functional capability of the function.
- **Minimal function**: - A function is said to be minimal if it is representing the function expression, if it is using minimal no of literals to represent the function.

- If a function is irredundant then, it may or may not be minimal, but if a function is minimal, then it is irredundant.
- $F(x, y, z) = xz' + y'z' + yz + xz$, this function is irredundant but not minimal
- $F(x, y, z) = X + yz + y'z'$
- Minimal $\subseteq$ irredundant

	xy	$x'y'$	$x'y$	xy	xy'
z		00	01	11	10
z'	00				
		0	2	6	4
z	01				
		1	3	7	5

Disadvantage: -

- If we deal with large no of variables (more than 4), then it is difficult to use k-map.

Break

Absorption law

- $a + ab = a$
- $a.(a+b) = a$

Compensation theorem

- $ab + a'c + bc = ab + a'c$
- $(a+b)(a' + c)(b + c) = (a+b)(a' + c)$

- $a + a'b = a + b$
- $a.(a'+b) = a.b$

Q The absorption law in Boolean algebra say that **(NET-JUNE-2011)**

(A) $X + X = X$

(B) $X \cdot X = X$

(C) $x + x \cdot y = x$

(D) None of the above

Q The number of min-terms after minimizing the following Boolean expression is _____ . **(GATE-2015) (1 Marks)**

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

(A) 1

(B) 2

(C) 3

(D) 4

Q Consider an expression & minimize it
 $a + a'b + a'b'c + a'b'c'd'$

Q Upto how many variables, can the Karnaugh map be used? **(NET-JUNE-2006)**

(A) 3

(B) 4

(C) 5

(D) 6

Q.54 Consider the following Boolean expression:

(GATE-2021)

$$F = (X + Y + Z)(\bar{X} + Y)(\bar{Y} + Z)$$

Which of the following Boolean expressions is/are equivalent to F ?

(a) $X\bar{Y} + Y\bar{Z} + \bar{X}\bar{Y}\bar{Z}$

(b) $(\bar{X} + \bar{Y} + \bar{Z})(X + \bar{Y})(Y + \bar{Z})$

(c) $(X + \bar{Z})(\bar{Y} + \bar{Z})$

(d) $X\bar{Y} + \bar{Z}$